

Project No. : 24167

Borehole Number : OW-31

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Project : MRT Orange Line (West Section)

Borehole Location : Tunnel (Sta. 15+200 EB)

Co-ordinate (N) : 1,521,329

Ground Elevation (m) : +101.200 m.MSL

Co-ordinate (E) : 663,700

G.W.L. (m) : 1.0 (below ground level)

Date Started : 25 Oct 2024

Total Depth (m) : 50.00

Date Finished : 29 Oct 2024

Depth (m)	SAMPLE			SUBSURFACE PROFILE		Atterberg Limits (%) 20 40 60 80	Unit Weight (kN/m³) 5 15	Suc (kN/m²) Δ								
	Symbol	Type	Number	Description	Symbol			SPT-N □ (Blows/300mm) □								
								20	40	60	80					
0				Ground Surface												
1				Fill Material / Top Soil	0.00											
2	■	ST	1													
3	■	ST	2	Very soft to soft CLAY, trace of sand, grey, high plasticity (CH)	2.50											
4																
5	■	ST	3													
6	■	ST	4													
7	■	ST	5													
8	■	ST	6													
9	■	ST	7													
10	■	ST	8	Medium stiff CLAY, grey, high plasticity (CH)	11.50											
11	■	ST	9													
12	■	ST	10	Stiff to very stiff CLAY, trace of sand, grey and greyish brown, high plasticity (CH)	13.00											
13	■	ST	11													
14	■	SS	1	Hard CLAY, trace of sand, greyish brown, high plasticity (CH)	14.50											
15	■	SS	2													
16	■	SS	3	Very stiff CLAY, grey, high plasticity (CH)	17.50											
17	■	SS	4													
18	■	SS	5	Hard silty CLAY, trace of sand, greyish brown, low plasticity (CL)	20.50											
19	■	SS	6													
20	■	SS	7	Very stiff silty CLAY, greyish brown, low plasticity (CL)	22.00											
21	■	SS	8													
22	■	SS	9	Hard silty CLAY, few to little sand, greyish brown and grey, low plasticity (CL)	23.50											
23	■	SS	10													
24	■	SS	11	Very dense silty SAND, grey and greyish brown, fine to medium grained (SM)	28.00											
25	■	SS	12													
26	■	SS	13													
27	■	SS	14													
28	■	SS	15													
29	■	SS	16													
30	■	SS	17	Hard silty CLAY, little sand for SS-19, grey, low plasticity (CL)	38.50											
31	■	SS	18													

Project : MRT Orange Line (West Section)

Depth (m)	SAMPLE			SUBSURFACE PROFILE		Atterberg Limits (%) 20 40 60 80	Unit Weight (kN/m³) 5 15	Suc (kN/m²) Δ 20 40 60 80 Δ			
	Symbol	Type	Number	Description	Symbol			SPT-N □ (Blows/300mm) □ 20 40 60 80			
40											
41	SS	19	Hard silty CLAY, little sand for SS-19, grey, low plasticity (CL)	41.50							85/225
42	SS	20	Very dense silty SAN, brown, fine grained (SM)								50/125
43											
44	SS	21									50/50
45	SS	22									50/100
46	SS	23									50/150
47	SS	23									
48	SS	24	Very dense SAND with silt, grey, fine to medium grained (SW-SM, SP-SM)	47.50							50/125
49											
50	SS	25									50/150
51			End of Borehole	50.00							
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Thin wall tube

Split spoon

Rock core

Wash

Auger

Atterberg Limits : PL Wn LL

Suc : Undrained shear strength from UC test

SPT-N : Field SPT-N (donut hammer with pulley and use split spoon sampling with liner)